

Postdoctoral Fellowship Application

NSF Atmospheric and Geospace Sciences Postdoctoral Research Fellowship

Department of Geography, University of Florida

Mostafa Rezaali

Ph.D. Candidate, Climate Science (Geography)

University of Florida

mostafarezaali@gmail.com | (352) 709-5350

References Cited

Lam, R., Sanchez-Gonzalez, A., Willson, M., Wyrnsberger, P., Fortunato, M., Alet, F., Ravuri, S., Ewalds, T., Eaton-Rosen, Z., Hu, W., Merose, A., Hoyer, S., Holland, G., Vinyals, O., Stott, J., Pritzel, A., Mohamed, S., and Battaglia, P. (2023). Learning skillful medium-range global weather forecasting. *Science*, 382(6677), 1416–1421.

Price, I., Sanchez-Gonzalez, A., Alet, F., Andersson, T. R., El-Kadi, A., Masters, D., Ewalds, T., Stott, J., Mohamed, S., Battaglia, P., Lam, R., and Willson, M. (2024). Probabilistic weather forecasting with machine learning. *Nature*, 637, 84–90.

Pegion, K., Kirtman, B. P., Becker, E., Collins, D. C., LaJoie, E., Burgman, R., Bell, R., DelSole, T., Min, D., Zhu, Y., Li, W., Sinsky, E., Guan, H., Gottschalck, J., Metzger, E. J., Barton, N. P., Achuthavarier, D., Marshak, J., Koster, R., Lin, H., Gagnon, N., Bell, M., Tippet, M. K., Robertson, A. W., Sun, S., Benjamin, S. G., Green, B. W., Bleck, R., and Kim, H. (2019). The Subseasonal Experiment (SubX): A multimodel subseasonal prediction experiment. *Bulletin of the American Meteorological Society*, 100(10), 2043–2060.

Vitart, F., Ardilouze, C., Bonet, A., Brookshaw, A., Chen, M., Codorean, C., Deque, M., Ferranti, L., Fucile, E., Fuentes, M., Hendon, H., Hodgson, J., Kang, H. S., Kumar, A., Lin, H., Liu, G., Liu, X., Malguzzi, P., Mallas, I., Manoussakis, M., Mastrangelo, D., MacLachlan, C., McLean, P., Minami, A., Mladek, R., Nakazawa, T., Najm, S., Nie, Y., Rixen, M., Robertson, A. W., Rutigliano, P., Sun, C., Takaya, Y., Tolstykh, M., Venuti, F., Waliser, D., Woolnough, S., Wu, T., Won, D. J., Xiao, H., Zaripov, R., and Zhang, L. (2017). The Sub-seasonal to Seasonal (S2S) Prediction Project Database. *Bulletin of the American Meteorological Society*, 98(1), 163–173.

White, C. J., Carlsen, H., Robertson, A. W., Klein, R. J. T., Lazo, J. K., Kumar, A., Vitart, F., Coughlan

- de Perez, E., Ray, A. J., Murray, V., Bharwani, S., MacLeod, D., James, R., Fleming, L., Morse, A. P., Eggen, B., Graham, R., Kjellstrom, E., Becker, E., Pegion, K. V., Holbrook, N. J., McEvoy, D., Depledge, M., Perkins-Kirkpatrick, S., Brown, T. J., Street, R., Jones, L., Remenyi, T. A., Hodgson-Johnston, I., Buontempo, C., Lamb, R., Meinke, H., Arheimer, B., and Zebiak, S. E. (2017). Potential applications of subseasonal-to-seasonal (S2S) predictions. *Meteorological Applications*, 24(3), 315–325.
- Otkin, J. A., Anderson, M. C., Hain, C., Svoboda, M., Johnson, D., Mueller, R., Tadesse, T., Wardlow, B., and Brown, J. (2016). Assessing the evolution of soil moisture and vegetation conditions during the 2012 United States flash drought. *Agricultural and Forest Meteorology*, 218–219, 230–242.
- Rasp, S., Dueben, P. D., Scher, S., Weyn, J. A., Mouatadid, S., and Thuerey, N. (2020). Weather-Bench: A benchmark data set for data-driven weather forecasting. *Journal of Advances in Modeling Earth Systems*, 12, e2020MS002203.
- Rezaali, M., Jahangir, M. S., Fouladi-Fard, R., and Keellings, D. (2024). An ensemble deep learning approach to spatiotemporal tropospheric ozone forecasting: A case study of Tehran, Iran. *Urban Climate*, 55, 101950.
- Rezaali, M., Quilty, J., and Karimi, A. (2021). Probabilistic urban water demand forecasting using wavelet-based machine learning models. *Journal of Hydrology*, 600, 126358.